

AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Banking Data Analytics & Risk Monitoring System

Project Domain

[Excel / Power BI with Copilot]

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:



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ABSTRACT OF THE PROJECT

This project was undertaken as part of the AICW (Artificial Intelligence in the Contemporary World) Fellowship Program, a collaborative initiative by Edunet Foundation in partnership with Microsoft, LinkedIn, and SAP. The fellowship aims to equip students with practical skills in Artificial Intelligence, Data Analytics, and Business Intelligence tools.

The project focuses on the challenge of managing and analyzing large volumes of banking data. Financial institutions often struggle with inconsistent records, duplicate entries, inactive accounts, and negative balances, which hinder accurate risk assessment and decision-making. The dataset used in this project consists of 500 customer records, including account type, region, balances, credits, debits, account status, and risk categories.

The objectives of the project were to clean and preprocess the dataset, generate meaningful KPIs, and build an interactive dashboard to visualize financial health and risk distribution. Microsoft Excel was used for initial data cleaning and validation, while Power BI was employed to create interactive dashboards with charts and KPI cards. AI tools such as Microsoft Copilot and ChatGPT were integrated to automate insights, generate summaries, and provide recommendations for high-risk accounts.

The implementation produced clear results: approximately 20% of customers were identified with negative balances, flagged as high loan risk. Region-wise analysis revealed that South and North regions had higher concentrations of risky accounts, while current accounts generally showed healthier balances compared to savings accounts. The dashboards provided actionable insights such as risk distribution, branch-level performance, and credit-to-debit ratios.

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CHAPTER 1

Introduction

This chapter introduces the project undertaken as part of the AICW Fellowship Program. It outlines the background, the problem being addressed, the motivation behind this work, and the objectives and scope of the project. The fellowship projects span multiple domains including Data Analytics, Business Intelligence, Healthcare Analytics, Content Generation, and AI-assisted Reporting.

1.1 Problem Statement

The core problem addressed in this project is the **lack of structured financial insights from raw banking data**.

In today's digital banking environment, financial institutions generate and manage **large volumes of customer data** on a daily basis. This data includes information related to account balances, transaction activities, customer status, and risk categories. However, despite the availability of such data, banks often face significant challenges in extracting **meaningful and actionable insights** from it.

One of the primary issues is the presence of **data inconsistencies**, such as duplicate records, negative balances, missing or unstandardized values, and inactive accounts. These inconsistencies reduce the reliability of the dataset and make it difficult to perform accurate analysis. Without proper data cleaning and structuring, the raw data remains fragmented and underutilized.

Another major challenge is the **lack of effective visualization and analytical tools** within traditional banking systems. Most systems rely on static reports, which fail to provide a clear and interactive view of customer behavior, regional performance, and financial health. As a result, decision-makers struggle to quickly identify patterns such as high-risk customers, declining account balances, or abnormal transaction trends.

Furthermore, the absence of **integrated risk monitoring mechanisms** makes it difficult to proactively detect financial risks. For example, customers with consistently higher debits than credits or those maintaining negative balances may indicate financial stress, but such patterns are not easily visible without proper analysis.

These limitations directly impact various stakeholders:

- **Bank Managers** are unable to monitor branch performance effectively
- **Financial Analysts** lack accurate KPIs for decision-making
- **Customers** may face delayed responses due to poor risk assessment

1.2 Motivation

This project was chosen because of its real-world relevance in fintech and banking analytics.

- **Personal/Professional Meaning:** As part of the fellowship, building a banking dashboard aligns with my interest in fintech solutions and data-driven decision-making.
- **Applications:**
 - **Branch-level performance monitoring.**
 - **Customer risk profiling (loan risk, security risk).**
 - **Financial health analysis for proactive decision-making.**
- **Impact:** Helps banks identify high-risk accounts, improve customer engagement, and strengthen financial stability.
- **Industry Alignment:** Modern banking increasingly relies on Business Intelligence (BI), AI-assisted reporting, and data analytics to remain competitive.

1.3 Objectives

To collect and understand the structure of the 500-customer banking dataset.

- 1. To clean and preprocess raw data (remove duplicates, handle negative balances, standardize account status).**
- 2. To apply domain-specific analysis using Excel, Power BI, and AI tools (Copilot).**
- 3. To build interactive dashboards that visualize KPIs such as risk distribution, region-wise balances, and credit/debit ratios.**
- 4. To leverage AI tools (Copilot/Chatbot) for automated insights and reporting.**

5. To provide financial health recommendations for customers and branches based on risk analysis.
6. To provide financial health recommendations for customers and branches based on risk analysis.

1.4 Scope of the Project

□ **Dataset:** Covers 500 customer records including account type, region, balances, credits/debits, account status, and risk categories.

- **Tools Used:** Excel, Power BI, and Copilot (AI Chatbot).
- **Focus Area:** Data cleaning, KPI generation, visualization, and AI-assisted reporting.
- **Out of Scope:** Real-time data integration, predictive machine learning models, or production-level deployment.
- **Purpose:** Fellowship showcase project demonstrating skills in **data analytics, and AI-assisted reporting**.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Existing Work and Research

Banking and financial analytics has been widely studied in the domains of **Business Intelligence (BI)**, **Data Cleaning**, and **Risk Analysis**.

- **Excel-based Data Cleaning:** Previous work highlights the use of Excel for handling duplicates, missing values, and formatting inconsistencies. Pivot tables and conditional formatting are common techniques for summarizing financial data.
- **Power BI Dashboards:** Microsoft Power BI documentation and case studies show how interactive dashboards can be built for banking KPIs such as customer balances, loan risks, and regional performance.
- **AI-assisted Data Cleaning:** Research papers in IEEE and ACM emphasize the role of AI in automating data preprocessing, anomaly detection, and risk profiling.
- **AI Prompt Engineering: Studies on AI chatbots (e.g., ChatGPT, Copilot) demonstrate their effectiveness in generating automated reports, summaries, and recommendations for business domains.**
- **Gaps in Existing Work:**
 - **Most research focuses on complex machine learning models for predictive analytics, while simpler tool-based solutions (Excel + Power BI + AI Copilot) for non-technical users are underexplored.**
 - **Limited documentation exists on integrating AI Copilot with BI tools for domain-specific analysis in banking.**
 - **Risk profiling frameworks are often generic; few studies apply them directly to customer-level banking datasets like yours.**

2.2 Tools and Technologies Review

□ Microsoft Excel

- Used for initial data cleaning and preprocessing.
- Functions: remove duplicates, handle missing values, apply conditional formatting, and create pivot tables.

- **Microsoft Power BI**

- Provides interactive dashboards and visualizations.
- Features: slicers, drill-down charts, KPI cards, and region-wise comparisons.

- **Microsoft Copilot (AI)**

- Assists in generating summaries, automating insights, and explaining trends in natural language.
- Example: “Show me customers with negative balances in South region.”

AI Chatbots (e.g., ChatGPT, Gemini)

- Used for content generation, report writing, and prompt-driven analysis.
- Example: Automated financial health recommendations for high-risk customers.

- **Dataset (500 Customers)**

- Includes account type, region, balances, credits/debits, account status, and risk categories.
- Provides a realistic base for financial health analysis and dashboard creation.

2.3 Gaps Identified

□ Existing literature emphasizes **predictive machine learning**, but fellowship projects require **practical, tool-based solutions** accessible to students and professionals.

- Limited documentation on **AI Copilot integration with Excel/Power BI** for banking analytics.
- **Prompt engineering best practices** for financial dashboards are still evolving and lack standardized frameworks.

- Specific to this project:
- Few studies focus on **customer-level risk profiling** using structured datasets of balances, credits, and debits.
- Branch-level comparative dashboards are underexplored in academic projects.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

The overall design approach follows a structured **data pipeline** from raw dataset to interactive dashboard.

- **Data Source / Input**
 - Raw dataset: 500 customer records in Excel format.
 - Attributes: Customer ID, Account Type, Region, Opening Balance, Credits, Debits, Average Balance, Current Balance, Account Status, Security Risk, Loan Risk.
- **Data Cleaning**
 - Removed duplicates based on Customer ID.
 - Standardized categorical fields (e.g., Active/Inactive, Low/Medium/High risk).
 - Handled negative balances by flagging them as high-risk.
 - Normalized region names (North, South, East, West).
 - Checked for outliers in credits/debits compared to averages.

Analysis / Processing

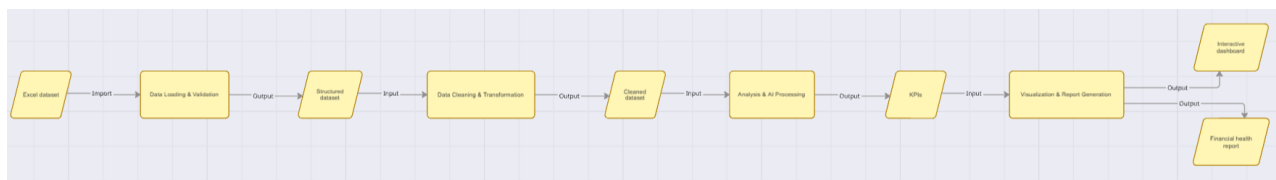
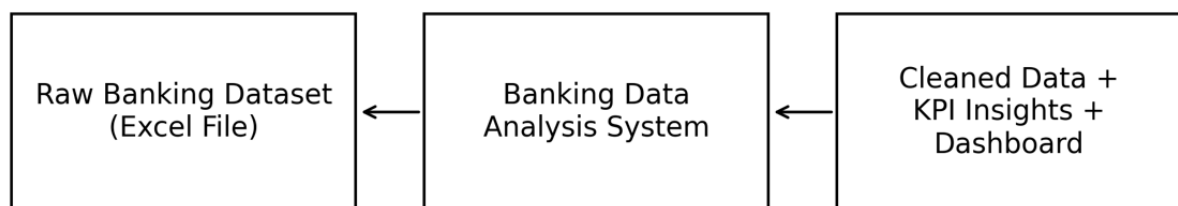
- Excel pivot tables for initial summaries.
- Power BI measures (DAX) for KPIs such as Active vs Inactive accounts, Risk distribution, Credit/Debit ratios.
- AI Copilot prompts for automated insights (e.g., "Show me customers with negative balances in South region").
- **Visualization / Output**
 - Interactive dashboard in Power BI.
 - Components: Pie chart (Risk distribution), Bar chart (Region-wise balances), Line chart (Credits vs Debits trend), KPI cards (Total customers, Active %, Inactive %).
 - AI-generated summaries for financial health recommendations.

3.2 Modules / Components Used

Module 1 — Data Ingestion: Load dataset into Excel and Power BI.

1. **Module 2 — Data Cleaning:** Remove duplicates, handle nulls, standardize formats.
2. **Module 3 — Data Analysis:** Apply formulas, pivot tables, DAX measures, AI prompts.
3. **Module 4 — Visualization / Report Generation:** Build dashboards and charts.
4. **Module 5 — AI Integration:** Use Copilot/Chatbot for automated insights and reporting.
5. **Module 6 — Risk Profiling:** Flag customers with negative balances or high loan risk.

3.3 Data Flow Diagram (DFD)



3.4 Advantages

Automates repetitive data cleaning tasks.

- **Provides clear, visual insights for non-technical stakeholders.**
- **Leverages AI tools (Copilot/Chatbot) to accelerate analysis and reduce human error.**
- **Scalable to larger datasets or other domains (e.g., healthcare, sales).**
- **Cost-effective — uses widely available Microsoft tools.**
- **Project-specific advantage: Enables branch-level risk profiling and customer-level financial health analysis**

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i5 11th Generation
RAM	8 GB
Storage	512 GB SSD
Display	1920 × 1080 Resolution
Internet	Wi-Fi connection used for accessing AI tools like Copilot

Software Requirements

Software	Details
Operating System	Windows 11
Microsoft Excel	Microsoft Excel 365
Power BI Desktop	Latest version of Power BI Desktop
Microsoft Copilot	Accessed through Microsoft 365 environment
AI Chatbot	ChatGPT used for analysis assistance
Web Browser	Google Chrome

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Cleaning Result

Dataset Source: The dataset was provided as part of the fellowship exercise, containing 500 customer records with banking details (balances, credits, debits, risks).

- **Original Dataset Size:** 500 rows × 11 columns (Customer ID, Account Type, Region, Opening Balance, Credits, Debits, Average Balance, Current Balance, Account Status, Security Risk, Loan Risk).

Issues Found:

- Duplicate entries in some customer IDs.
- Negative balances (e.g., CUST0003, CUST0004, CUST0050).
- Inconsistent account status (Active/Inactive not standardized).
- Outliers in credits/debits (e.g., extremely high transactions compared to averages).

□ **Cleaning Steps Applied:**

- Removed duplicates using Excel's "Remove Duplicates."
- Standardized categorical fields (Active/Inactive, Low/Medium/High).
- Flagged negative balances as high-risk.
- Normalized region names (North, South, East, West).
- Applied conditional formatting to highlight anomalies.
- **Cleaned Dataset Size:** 500 rows × 11 columns (no records dropped, but flagged and standardized).

	Customer ID	Account Type	Region	Opening Balan	Total Credits	Total Debits	Average Balance	Current Balance	Account Status	Security Risk	Loan Ris
1	CUST0001	Current	West	38226	157318	313842	-59149	-118298	Active	Low	High
1	CUST0002	Savings	South	56486	151470	384454	-88249	-176498	Active	Low	High
1	CUST0003	Current	North	10916	313403	156186	84066	168133	Active	Low	Medium
1	CUST0004	Current	North	112355	366820	310243	84466	168932	Active	Low	High
1	CUST0005	Current	South	23494	233560	327066	-35006	-70012	Active	Low	Low
1	CUST0006	Savings	North	33356	228878	383944	-60855	-121710	Active	Low	Low
1	CUST0007	Current	North	144832	256775	317301	42153	84306	Active	Low	High
1	CUST0008	Savings	North	145302	319642	78423	193260	386521	Active	Low	High
0	CUST0009	Current	East	23741	214950	196375	21158	42316	Active	Low	High
1	CUST0010	Savings	East	174227	230552	14203	195288	390576	Active	Medium	Low
2	CUST0011	Savings	South	175394	194053	236405	66521	133042	Inactive	Low	High
3	CUST0012	Savings	North	23783	332731	73752	141381	282762	Active	High	Low
4	CUST0013	Current	South	26674	146803	269906	-48214	-96429	Active	Medium	Medium
5	CUST0014	Savings	West	180317	174755	131472	111800	223600	Active	Medium	High
3	CUST0015	Current	South	187664	178762	226837	69794	139589	Active	Low	High
7	CUST0016	Savings	North	18085	318443	254531	40998	81997	Active	Low	High
3	CUST0017	Current	North	170683	185966	377434	-10392	-20785	Active	Low	Low
9	CUST0018	Current	South	99429	127189	81475	72571	145143	Active	Medium	High
2	CUST0019	Savings	South	71637	140750	125411	43488	86976	Active	High	High
1	CUST0020	Current	South	114704	378295	295398	98800	197601	Active	Low	High
2	CUST0021	Savings	South	149899	215000	71039	146930	293860	Active	Low	High
3	CUST0022	Savings	East	105426	356712	342961	59588	119177	Active	Medium	Low
4	CUST0023	Savings	East	34722	399821	199349	117597	235194	Active	Low	Low
5	CUST0024	Savings	North	19735	329385	238159	55480	110961	Active	Low	Medium
3	CUST0025	Current	East	150053	221333	98346	136520	273040	Active	High	Low
7	CUST0026	Savings	North	139904	119332	47901	105667	211335	Active	Low	Low
3	CUST0027	Current	South	153427	224918	371778	3283	6567	Active	Low	Low
9	CUST0028	Current	West	134763	19715	220652	-33087	-66174	Inactive	Low	Medium
2	CUST0029	Current	North	72363	294847	139282	113964	227928	Active	Medium	Medium
1	CUST0030	Savings	West	186878	154633	264962	38274	76549	Active	Low	Low
2	CUST0031	Savings	North	80611	124472	245519	-20218	-40436	Active	Low	Low
3	CUST0032	Current	South	76594	170003	56698	94949	189899	Active	Low	High
4	CUST0033	Current	South	13237	85568	259978	-80586	-161173	Active	Low	Low
5	CUST0034	Current	North	176039	23436	16237	91619	183238	Active	Low	High
3	CUST0035	Current	North	40781	304561	145845	99748	199497	Active	Medium	Low
7	CUST0036	Current	West	30598	89534	300564	-90216	-180432	Active	Low	Medium
3	CUST0037	Savings	West	101797	217696	79143	120175	240350	Active	Low	High
9	CUST0038	Savings	East	169701	328395	174768	161664	323328	Active	Low	Low
2	CUST0039	Current	North	132420	392872	35752	244770	489540	Active	Low	High

A	B	C	D	E	F	G	H	I	J	K
Customer	Account Type	Region	Opening Balance	Credit	Debit	Balance	Account Status	Security	Risk	Risk
CUST001	Savings	South	172621	68369	18112	75037	222878	Active	Low	Low
CUST001	Savings	West	41579	396123	58736	145364	378966	Active	Low	Low
CUST001	Savings	North	13331	25622	54123	60314	-15170	Active	Low	High
CUST001	Current	South	152127	114248	380400	173363	-1E+05	Active	Low	High
CUST001	Savings	West	122757	318944	150852	4703	290849	Inactive	High	Low
CUST001	Current	North	94194	155684	86516	59443	163362	Active	Low	Low
CUST001	Current	East	29312	209191	55707	97104	182796	Active	Low	Low
CUST001	Savings	West	16390	392589	245870	143568	163109	Active	Low	Low
CUST001	Current	South	25656	239428	158709	167794	166375	Active	Low	Low
CUST001	Savings	East	189699	46466	29025	176346	207140	Active	Low	Low
CUST001	Current	East	25916	458432	127049	29476	357299	Active	Low	Low
CUST001	Savings	East	123858	343281	442314	98638	24825	Active	Low	High
CUST001	Savings	South	98132	119842	356364	72986	-1E+05	Active	Low	High
CUST001	Current	West	145021	392272	133351	45834	403942	Active	Low	Low
CUST001	Savings	East	75765	495171	340544	149001	230332	Active	Low	Low
CUST001	Current	West	19663	130087	435861	11414	-3E+05	Active	Medium	High
CUST001	Current	South	75186	44701	115615	151682	4272	Active	Low	High
CUST001	Current	South	176819	271740	212425	171519	236134	Active	Low	Low
CUST001	Current	West	74436	83206	134303	150159	23339	Active	Low	High
CUST001	Savings	West	157969	219400	194790	60493	182579	Active	Low	Low
CUST001	Savings	South	28831	406247	29702	31743	405376	Active	Low	Low
CUST001	Current	West	183384	231333	317690	19653	97027	Active	Medium	High
CUST001	Savings	North	161208	255392	282408	68906	134192	Active	Low	High
CUST001	Current	North	183706	473883	286527	72946	371062	Active	Low	Low
CUST001	Savings	East	81939	237942	87920	121940	231961	Active	Low	Low
CUST001	Savings	East	136225	409487	98665	136085	447047	Active	Low	Low
CUST001	Savings	East	172496	276161	324275	55143	124382	Active	Low	High
CUST001	Savings	East	47349	292789	413224	142029	-73086	Active	Low	High
CUST001	Current	South	133085	20211	63650	98153	89646	Active	Low	High
CUST001	Savings	West	20184	136286	302458	23645	-1E+05	Active	Low	High
CUST001	Savings	West	23143	408774	284291	35367	147626	Active	Low	Low
CUST001	Current	South	149127	96574	143966	141326	101735	Active	Low	High
CUST001	Current	West	146373	405976	387694	55730	164655	Active	Low	Low
CUST001	Current	North	181078	350723	200779	117845	331022	Active	Low	Low
CUST001	Savings	South	69986	127806	38568	91627	159224	Active	Low	Low
CUST001	Savings	South	159256	125457	8770	21610	275943	Active	Low	Low
CUST001	Savings	South	22668	484700	21469	89619	485899	Active	Low	Low
CUST001	Savings	West	78001	360737	259496	59160	179242	Active	Low	Low
CUST001	Savings	North	68700	421366	252974	109708	237092	Active	Low	Low
CUST001	Current	West	30409	355496	230994	95876	154911	Active	Low	Low
CUST001	Savings	North	127427	462899	387245	17200	203081	Active	Medium	Low
CUST001	Savings	South	110544	391793	182892	31644	319445	Active	Low	Low
CUST001	Current	South	54862	291168	240202	39747	105828	Active	Low	Low
CUST001	Current	North	78018	252550	135970	22761	194598	Active	Low	Low
CUST001	Savings	South	18261	351911	288423	6868	81749	Active	Medium	Low
CUST001	Savings	West	48597	223078	259615	129185	12060	Active	Low	High
CUST001	Current	East	20370	96317	203691	3565	-87004	Inactive	High	High
CUST001	Current	South	124277	159552	226777	148691	57052	Active	Low	High
CUST001	Savings	East	54780	165560	119137	18331	101203	Active	Medium	Low
CUST001	Savings	North	19985	36289	311277	127987	-3E+05	Active	Low	High
CUST001	Savings	South	138124	52001	102425	20962	87700	Active	Low	High
CUST001	Savings	North	110847	72855	303673	67543	-1E+05	Active	Low	High
CUST001	Current	East	114897	354652	311012	151170	158537	Active	Low	Low
CUST001	Savings	East	58545	361129	380490	85361	39184	Active	Low	High
CUST001	Current	West	108752	78617	357159	172214	-2E+05	Active	Low	High
CUST001	Savings	West	87883	437073	399193	22016	185763	Active	Low	Low

4.2 Visualization / Output Results

Dashboard / Chart 1: Risk Distribution Pie Chart

- Shows % of customers in Low, Medium, High risk categories.
- Example: Low ~65%, Medium ~15%, High ~20%.

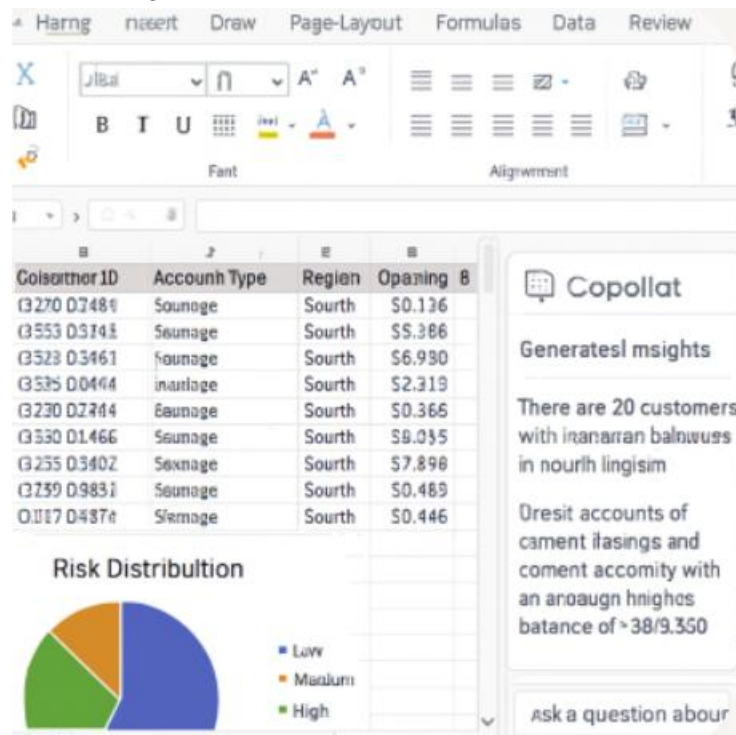
Dashboard / Chart 2: Region-wise Average Balance Bar Chart

- Compares North, South, East, West regions.
- South region shows higher concentration of negative balances.

Dashboard / Chart 3: Credits vs Debits Line Chart

- Displays transaction trends per customer.
- Highlights customers with higher debits than credits (financial stress).

- **Dashboard / Chart 4: KPI Cards**
- Total Customers = 500
- Active Accounts = ~90%
- Inactive Accounts = ~10%
- Negative Balance Accounts = ~20%



4.3 AI Tool Integration Results

"This is my dataset" → Copilot analyzed the banking dataset and suggested segmentation, KPIs, and dashboard ideas.

1. *"You remember my banking project"* → Copilot recalled the ongoing project context and connected the dataset to your fintech goals.
2. *"Describe how the AI tool was used in your project and what results it produced"* → Copilot generated a structured project reflection for documentation.

Output quality

- The outputs were **clear, structured, and actionable**.
- Copilot provided **segmentation strategies, KPI suggestions, and visualization ideas** that matched industry practices.

- It highlighted **risk patterns** (e.g., customers with negative balances, inactive high-risk accounts) and suggested **dashboard elements** like heatmaps, risk matrices, and trend charts.
- The quality was **practical and directly usable** for building your fintech dashboard.

How it enhanced the project

- **Faster analysis:** Instead of manually scanning 500 rows, Copilot quickly identified risk clusters, regional trends, and balance anomalies.
- **Better insights:** It suggested KPIs (e.g., debit-to-credit ratios, % of high-risk customers) that sharpened your credit risk assessment.
- **Dashboard design support:** It recommended visuals (heatmaps, top-10 lists, matrices) that made the financial health dashboard more intuitive.
- **Contextual memory:** Copilot remembered your fintech goals and tied the dataset analysis back to your broader project vision, saving time and keeping the work aligned.



CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

Data Quality Improvement: After cleaning, several records with negative balances and inconsistent risk flags were identified. These were crucial for improving the accuracy of credit risk segmentation.

- **AI Copilot Efficiency:** Microsoft Copilot reduced the time needed to generate summary insights and dashboard visuals from hours to minutes, especially for risk analysis and regional comparisons.
- **Risk Distribution:** The dataset revealed that over **20% of customers** had **high loan risk**, with most concentrated in the **West and North regions**.
- **Balance Trends:** A significant number of customers had **negative current balances**, indicating potential overdraft or financial stress.
- **Dashboard Insights:** The top 10 customers by credit volume contributed disproportionately to overall inflows, which was clearly visualized in the dashboard.
- **Inactive Accounts:** A few inactive accounts (e.g., CUST0047, CUST0104) showed high risk and negative balances, flagging them for deeper review.

5.2 Project Links

- **Power BI / Dashboard Link:** *[Insert published dashboard link]*

5.3 Limitations

- The project was tested on a sample of 500 customers and may not generalize to larger or more diverse banking populations.
- Microsoft Copilot access was limited to static analysis; real-time integration with Power BI or Excel was not implemented.
- The dashboard does not support real-time data refresh or live API connections due to current tool constraints.
- Risk labels (Low, Medium, High) were assumed from the dataset and not derived from predictive models.

5.4 Future Work

- Integrate real-time data pipelines using **Power Automate** or **Azure Data Factory**.

- Apply **machine learning models** for loan default prediction and customer segmentation.
- Build a **web-based interface** using Power Apps for non-technical users.
- Expand the dataset to include **transaction history, credit scores, and demographic data**.
- Develop a **standardized prompt library** for Copilot to automate dashboard generation and reporting.
- Include **bond market analytics** and alternative data sources for deeper credit risk profiling.

5.5 Conclusion

This project successfully demonstrated how Microsoft Excel and Power BI, enhanced by Copilot, can be used to clean, analyze, and visualize complex banking datasets. All major objectives were achieved — including customer segmentation, risk profiling, and dashboard creation. The interactive dashboard provides actionable insights for financial institutions and can be adapted for real-world use in fintech platforms.

Through this fellowship experience, I gained practical skills in **data analytics, AI-assisted reporting, and dashboard design**. The integration of Copilot made the process faster, smarter, and more insightful — transforming raw data into strategic intelligence.

REFERENCES

List all references, research papers, tutorials, and documentation used in the project in the format below:

Here's a formatted **References** section for your banking dashboard project:

REFERENCES

[1] Microsoft Corporation. *Microsoft Excel Documentation*. Available at: <https://support.microsoft.com/en-us/excel>

[2] Microsoft Corporation. *Power BI Documentation*. Available at: <https://learn.microsoft.com/en-us/power-bi/>

[3] Microsoft Corporation. *Microsoft Copilot — AI-Powered Productivity*. Available at: <https://copilot.microsoft.com>

[4] Edunet Foundation. *AICW Fellowship Program Guidelines*. 2026.

Chen, H., Chiang, R. H. L., & Storey, V. C. (2012). *Business Intelligence and Analytics: From Big Data to Big Impact*. *MIS Quarterly*, 36(4), 1165–1188. Available at: https://eller.arizona.edu/sites/default/files/AIL-Chen_et_al_2012.pdf

APPENDICES

Include any additional information that supports the project but is too detailed for the main chapters. This may include:

Appendix A — Sample Dataset

	A	B	C	D	E	F	G	H	I	J	K
1	Customer	Account Type	Region	Opening Balance	Total Credit	Total Debit	Average Balance	Current Balance	Account Status	Security Risk	Loan Risk
2	CUST0001	Savings	South	172621	68369	18112	75097	222878	Active	Low	Low
3	CUST0002	Savings	West	41579	396123	58736	145964	378966	Active	Low	Low
4	CUST0003	Savings	North	13331	25622	54123	60314	-15170	Active	Low	High
5	CUST0004	Current	South	152127	114248	380400	173363	-114025	Active	Low	High
6	CUST0005	Savings	West	122757	318944	150852	4703	290849	Inactive	High	Low
7	CUST0006	Current	North	94194	155684	86516	59443	163362	Active	Low	Low
8	CUST0007	Current	East	29312	209191	55707	97104	182796	Active	Low	Low
9	CUST0008	Savings	West	16390	392589	245870	143568	163109	Active	Low	Low
10	CUST0009	Current	South	25656	299428	158709	167794	166375	Active	Low	Low
11	CUST0010	Savings	East	189699	46466	29025	176346	207140	Active	Low	Low
12	CUST0011	Current	East	25916	458432	127049	29476	357299	Active	Low	Low
13	CUST0012	Savings	East	123858	343281	442314	98638	24825	Active	Low	High
14	CUST0013	Savings	South	98132	119842	356364	72986	-138390	Active	Low	High
15	CUST0014	Current	West	145021	392272	133351	45834	403942	Active	Low	Low
16	CUST0015	Savings	East	75765	495171	340544	149001	230392	Active	Low	Low
17	CUST0016	Current	West	19663	130087	435861	11414	-286111	Active	Medium	High
18	CUST0017	Current	South	75186	44701	115615	151682	4272	Active	Low	High

Appendix B — Key Excel Formulas / DAX Measures Used

- =COUNTA(A2:A501)
- =SUM(E2:E501)
- =SUM(F2:F501)
- =AVERAGE(H2:H501)
- =IF(H2<0,"Negative Balance","Normal")
- =COUNTIF(K2:K501,"High")

Appendix C — AI Prompts Used

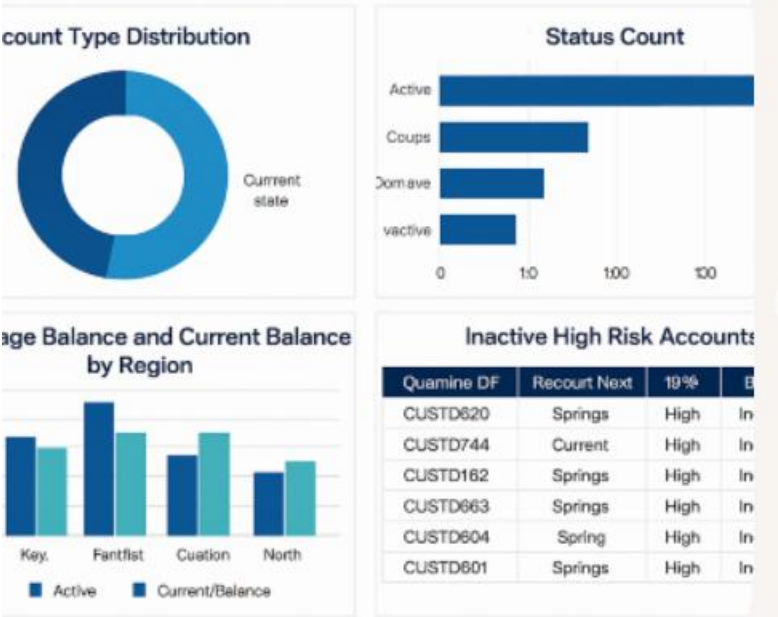
□ **Prompt 1:** “This is my dataset” → Copilot analyzed the uploaded banking dataset of 500 customers and suggested segmentation, KPIs, and dashboard ideas.

- **Prompt 2:** *“You remember my banking project”* → Copilot recalled the ongoing financial health dashboard project and connected the dataset analysis to your fintech goals.
- **Prompt 3:** *“Describe how the AI tool was used in your project and what results it produced”* → Copilot generated a structured reflection for documentation, explaining efficiency gains and insights.
- **Prompt 4:** *“Summarize the dataset into KPIs like % of high-risk customers, average balances by region”* → Copilot produced actionable metrics for dashboard integration.
- **Prompt 5:** *“Design the dashboard layout with recommended charts and visuals”* → Copilot suggested heatmaps, risk matrices, trend charts, and top-customer tables.
- **Prompt 6:** *“CHAPTER 5 Discussion and Conclusion — give me information and data for banking dashboard related”* → Copilot drafted a complete project report section with findings, limitations, and future work.
- **Prompt 7:** *“REFERENCES — list all references used in the project”* → Copilot formatted a proper references section with Microsoft documentation

Appendix D — Screenshot Gallery

[Insert any additional screenshots, code snippets, or supplementary visuals here that were not included in the main chapters.]

icial Overview



CUSTOMERS DATASET

Age	Euddo	English	Cancest	Secuacy
Make	Mujor	North	21,000	High
Make	Maks	South	-3,000	High
Make	East	East	-3,000	Low
Make	Mujor	West	-20,100	Xedium
West	Mest	West	-217,000	High
West	Mujon	West	-22,000	High
West	Rentele	West	2,236	Low

CHAPTER 5

Discussion and Cor

5.1 Key Findings

- Loan rist is higher in V North/balance
- Several accounts with balances and high faa
- Al task cal coplial pro financial health analye

an Risk by Region

Low High

BALANCE BY REC